

COE 1195: Advanced Digital Design

Lab 4 – 32-bits MIPS CPU

Milestone 4

Dr. Amr Mahmoud

Milestone 4

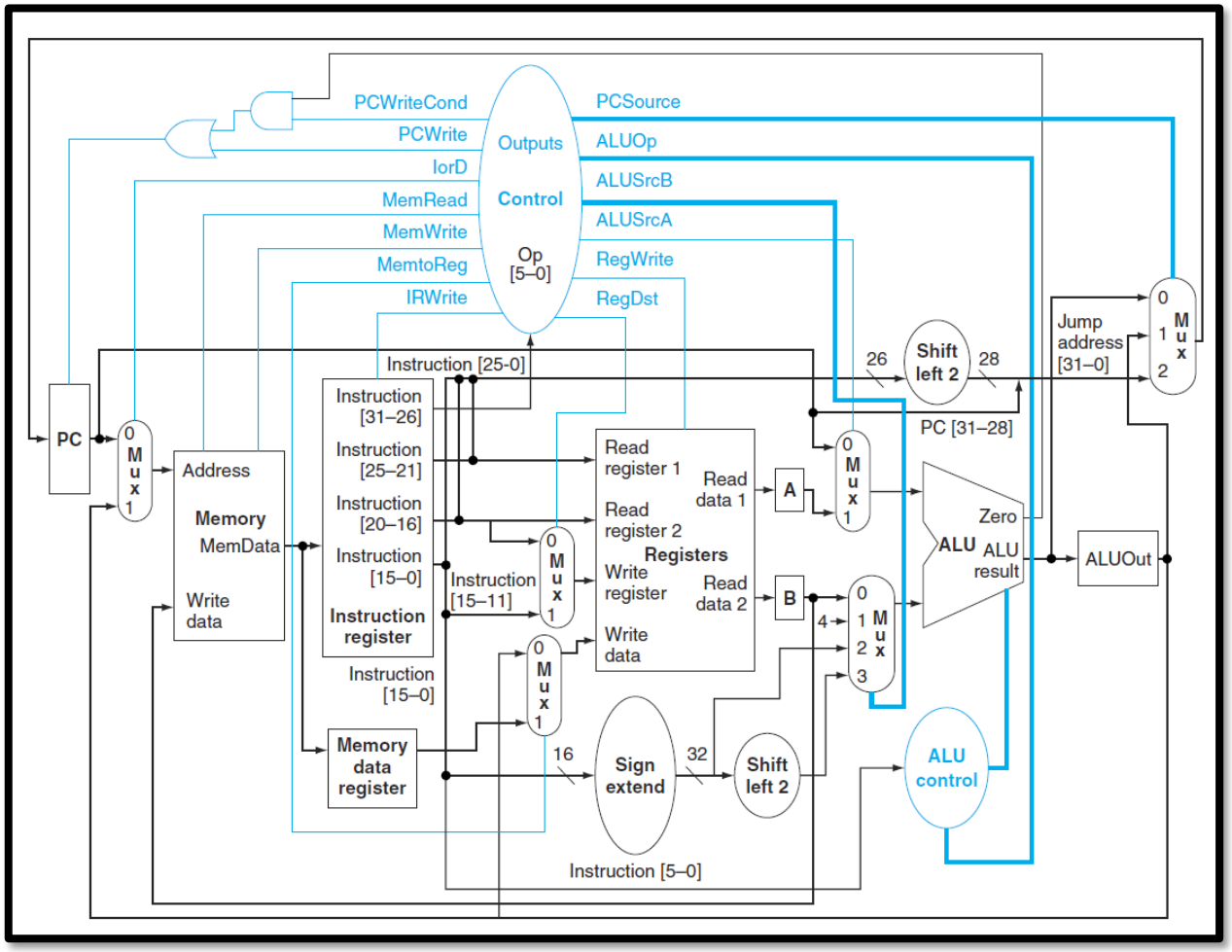
1. Implement Memory-Type instructions (Store/Load)

2. Implement CLO Instruction

Mem-Type Instructions

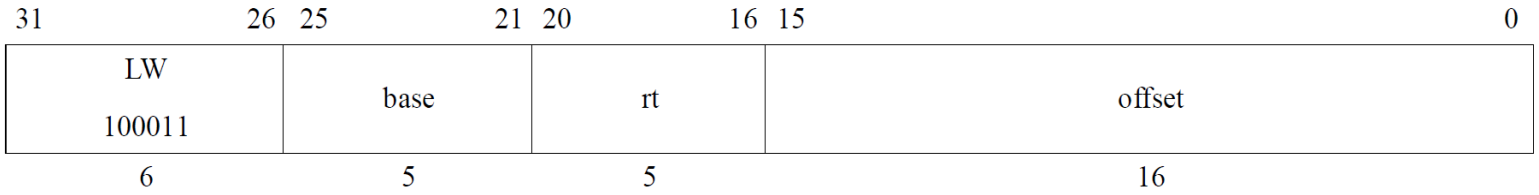
ADDI	ADDU	AND
ORI	SUB	SRA
SLLV	SLL	SLTI
LUI	SW	LW
LH	CLO	BNE
J	JR	MULTU
MFLO	MFHI	

Milestone 4 – Implement Mem-Type instructions



MIPS Multicycle CPU Datapath

LW



Format: `LW rt, offset(base)` MIPS32 (MIPS I)

Purpose:

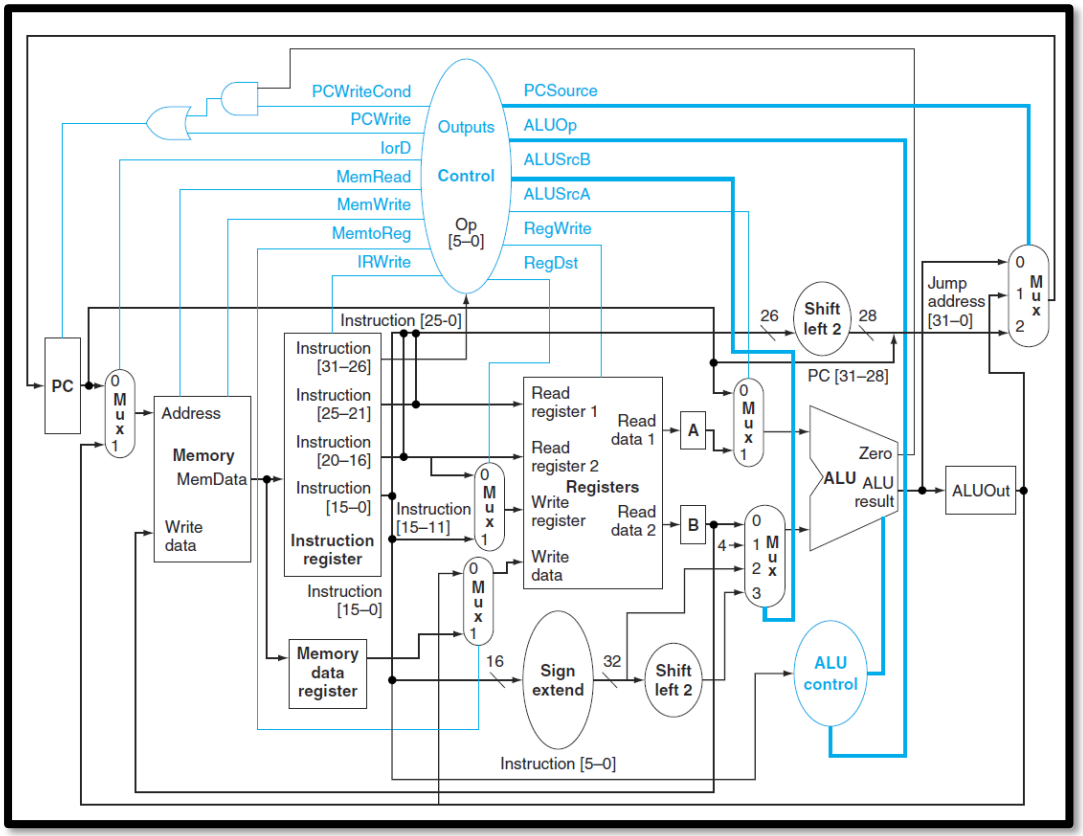
To load a word from memory as a signed value

Description: `rt ← memory[base+offset]`

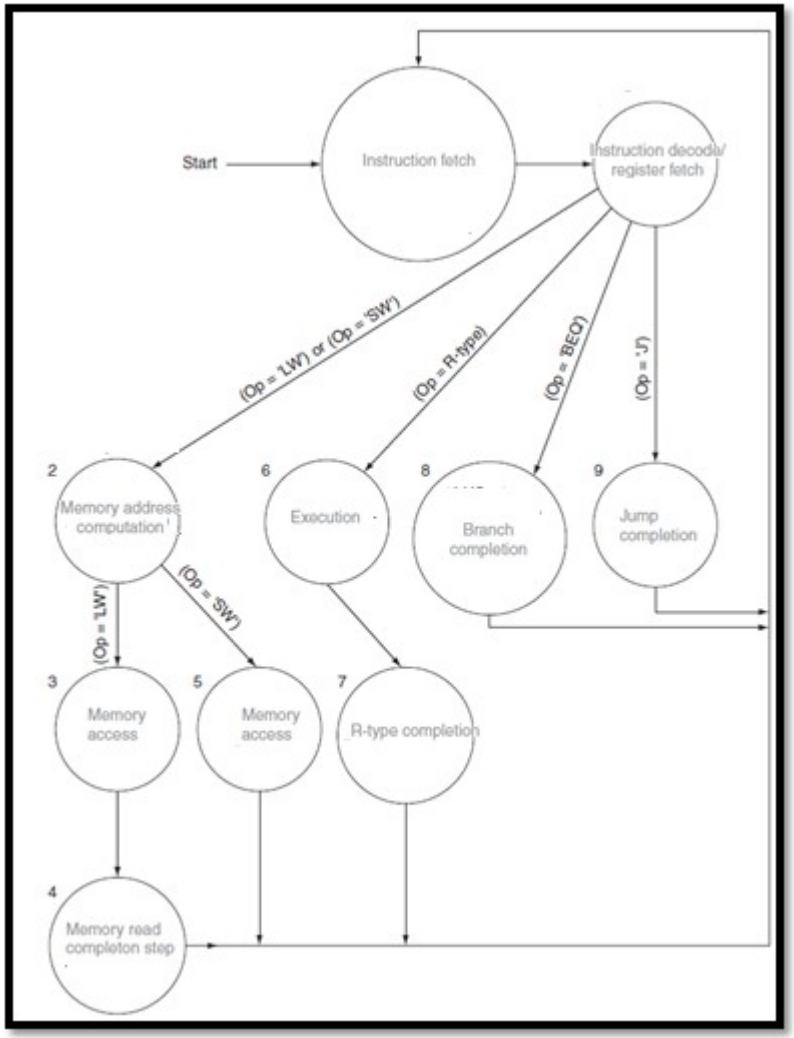
The contents of the 32-bit word at the memory location specified by the aligned effective address are fetched, sign-extended to the GPR register length if necessary, and placed in GPR *rt*. The 16-bit signed *offset* is added to the contents of GPR *base* to form the effective address.

Datapath and Control

MIPS Multicycle CPU FSM



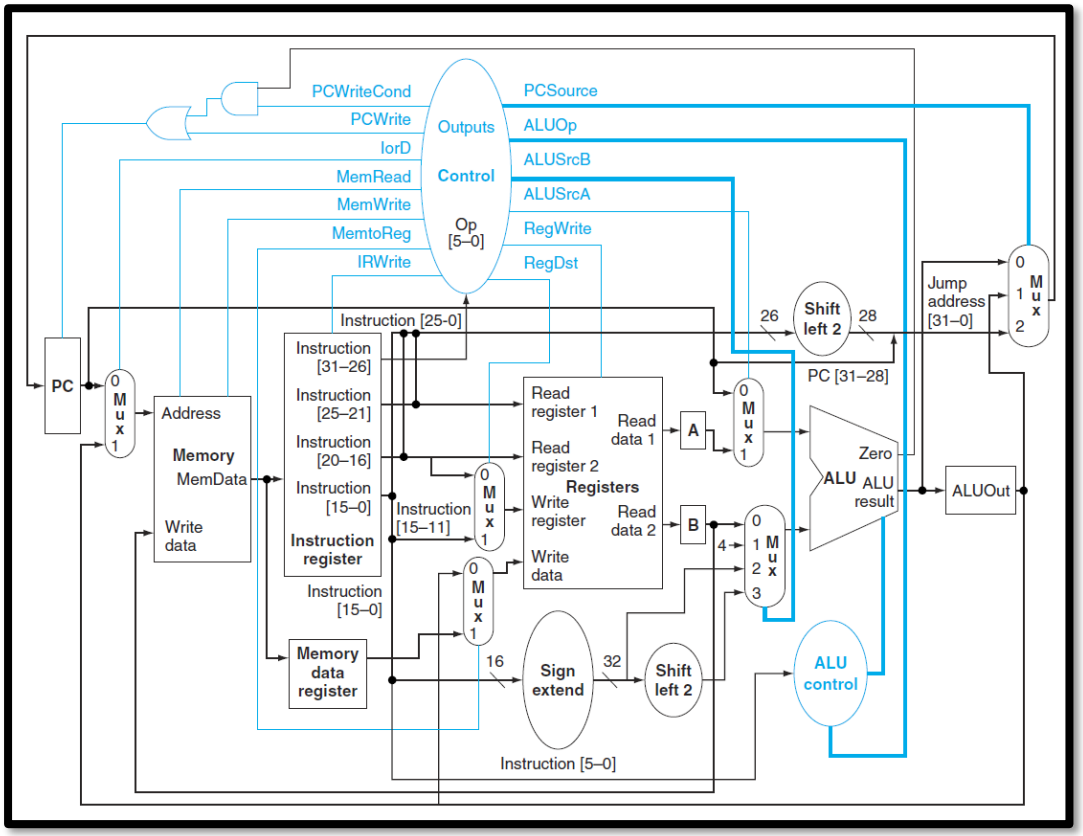
MIPS Multicycle CPU Datapath



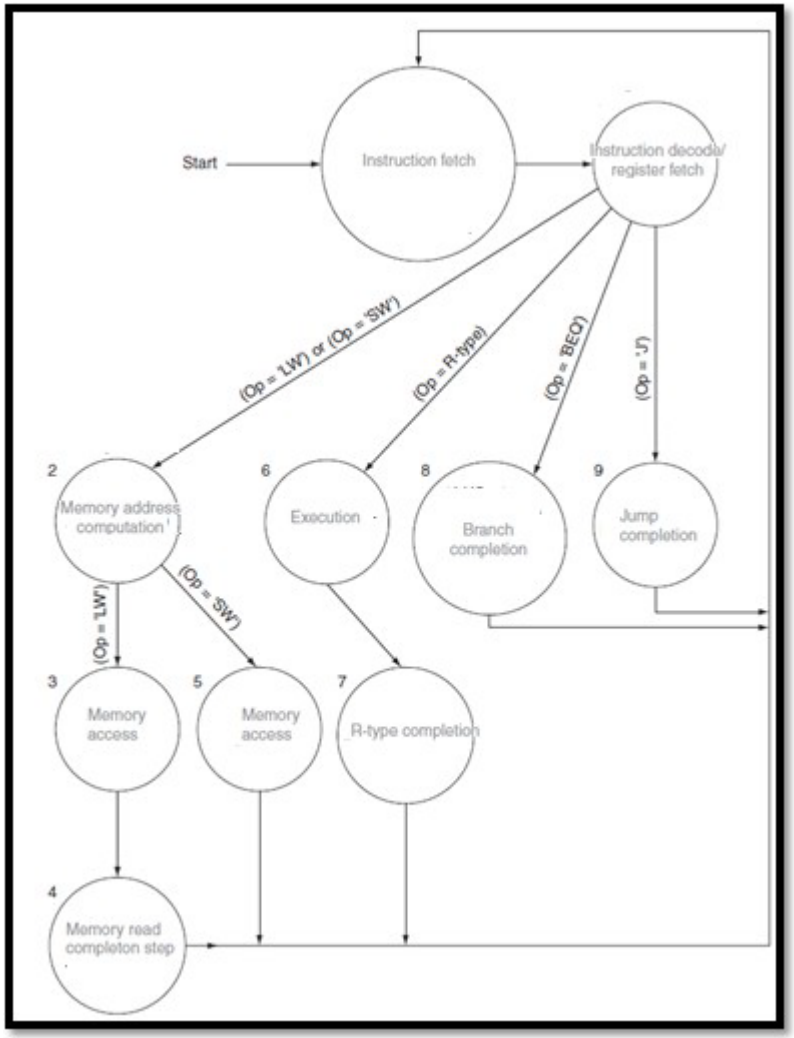
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



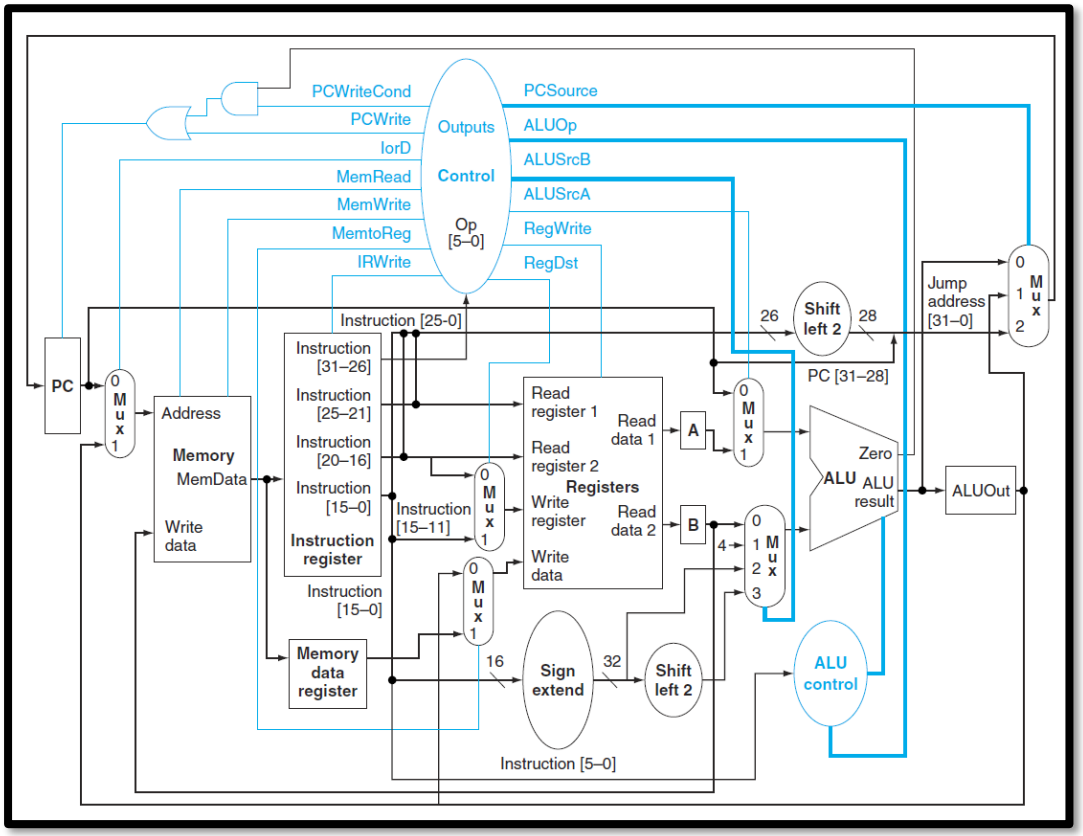
MIPS Multicycle CPU Datapath



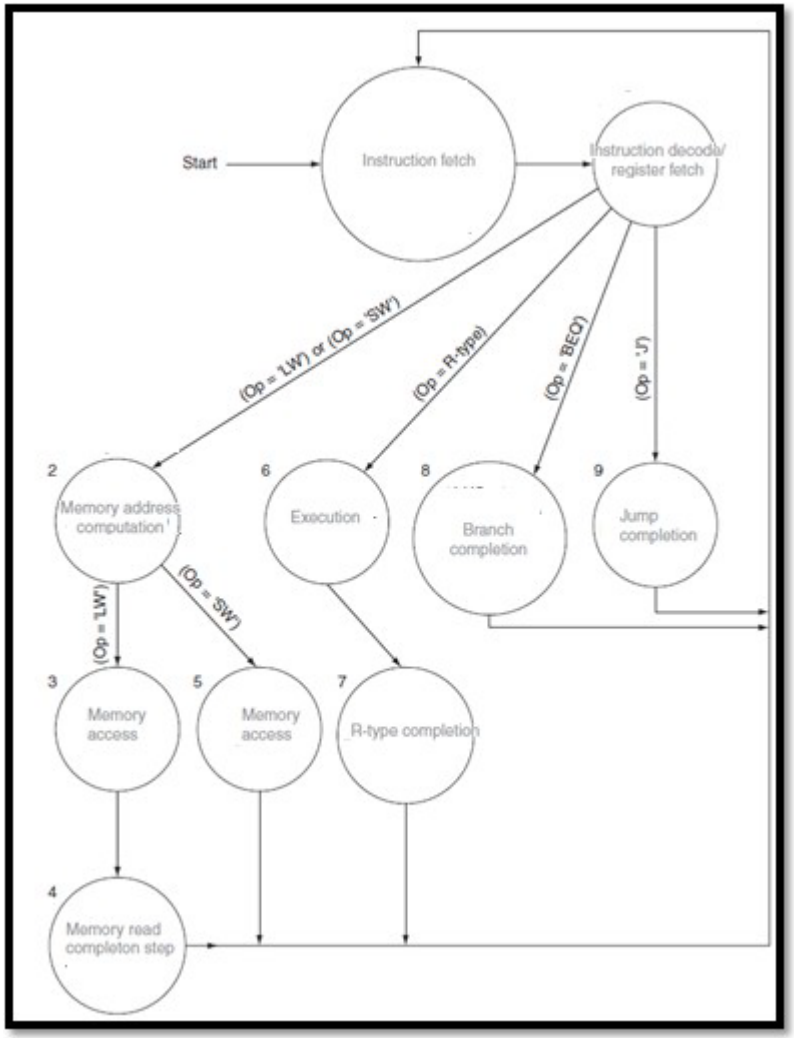
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



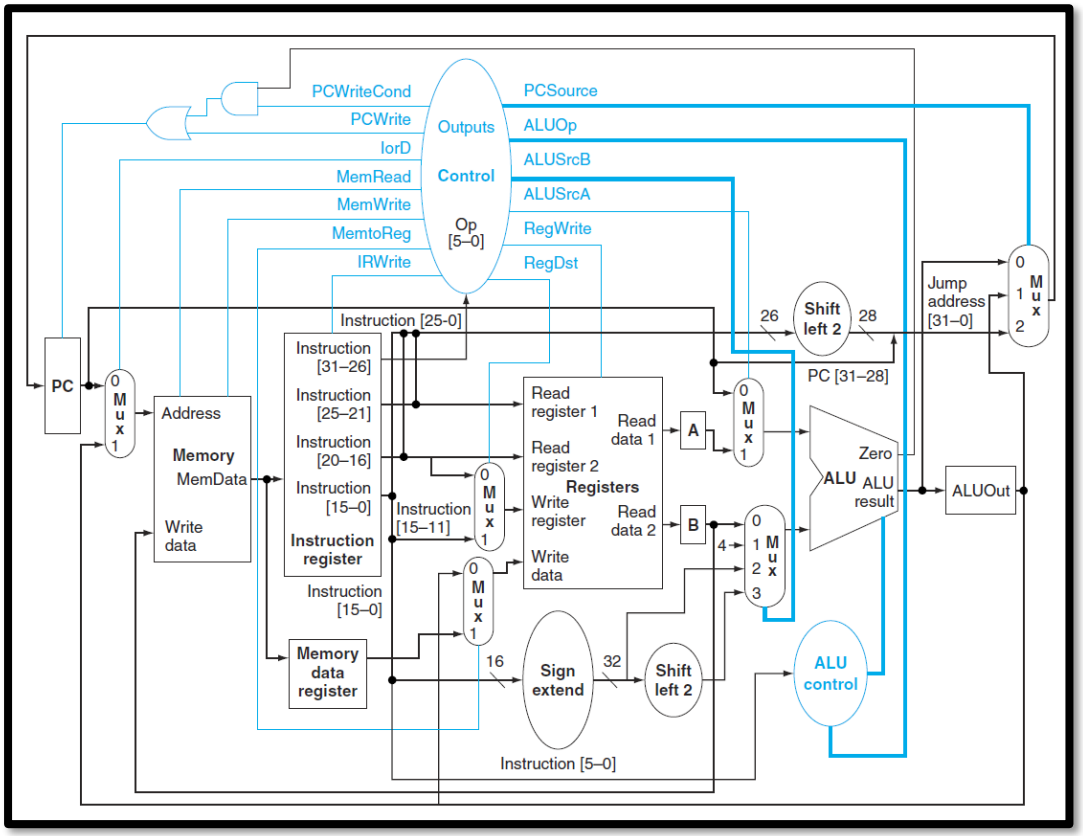
MIPS Multicycle CPU Datapath



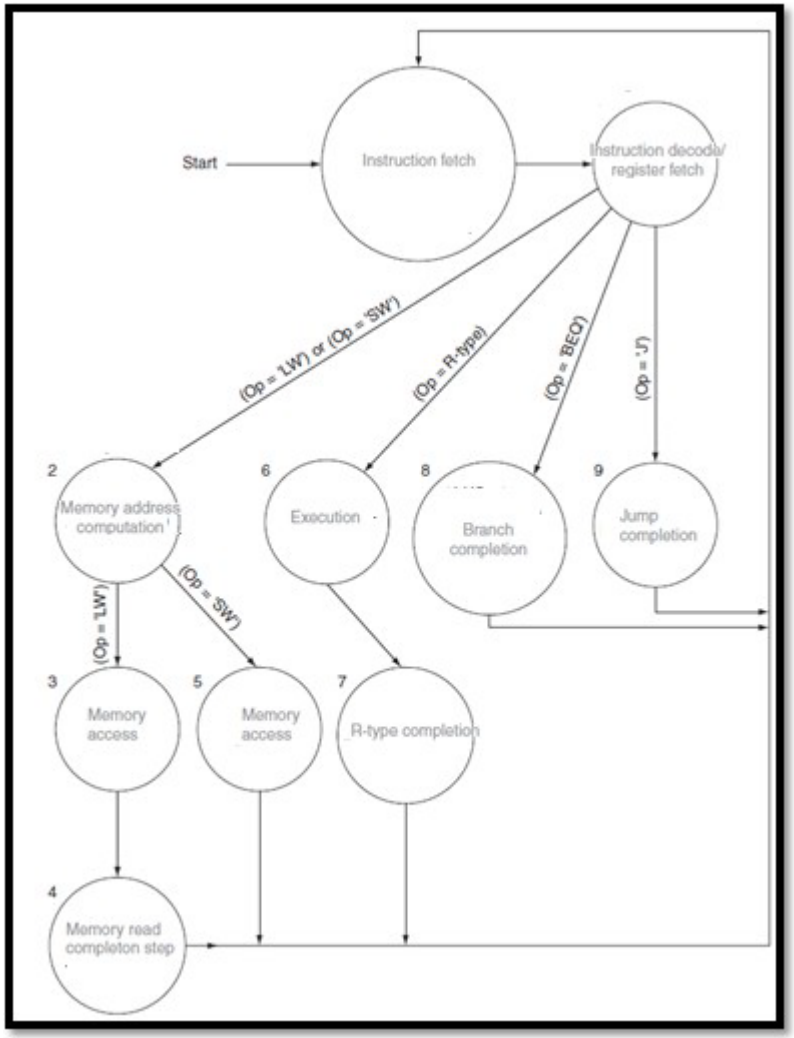
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM

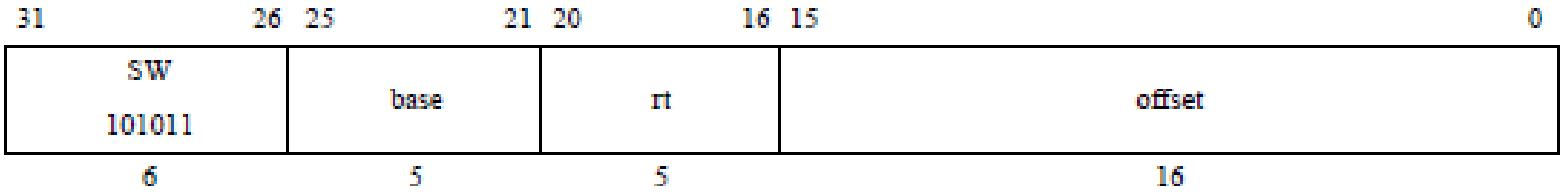


MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller

SW



Format: `SW rt, offset(base)` MIPS32 (MIPS I)

Purpose:

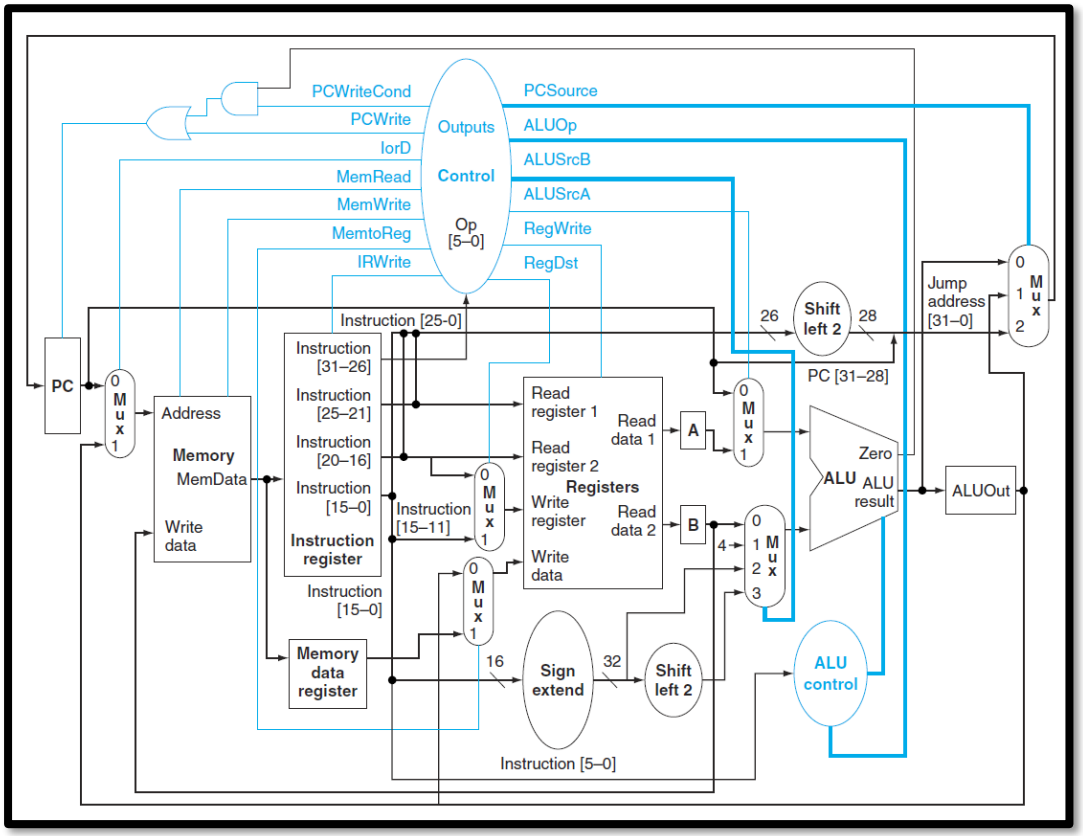
To store a word to memory

Description: `memory[base+offset] ← rt`

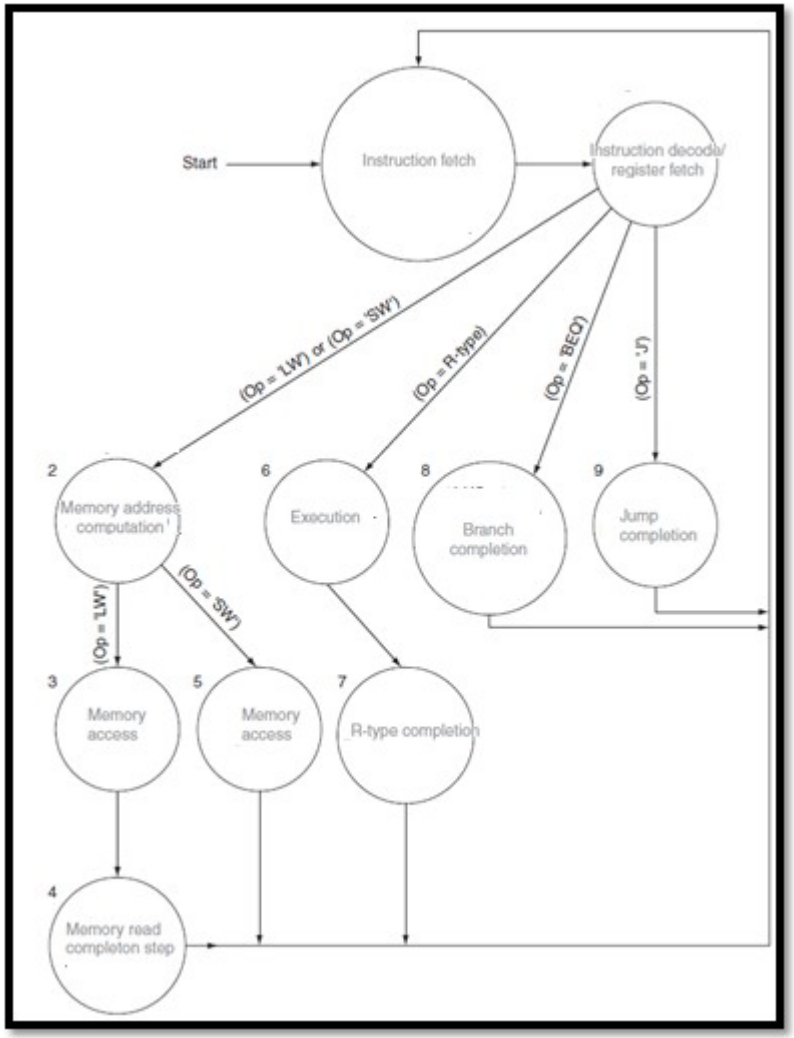
The least-significant 32-bit word of register *rt* is stored in memory at the location specified by the aligned effective address. The 16-bit signed *offset* is added to the contents of GPR *base* to form the effective address.

Datapath and Control

MIPS Multicycle CPU FSM



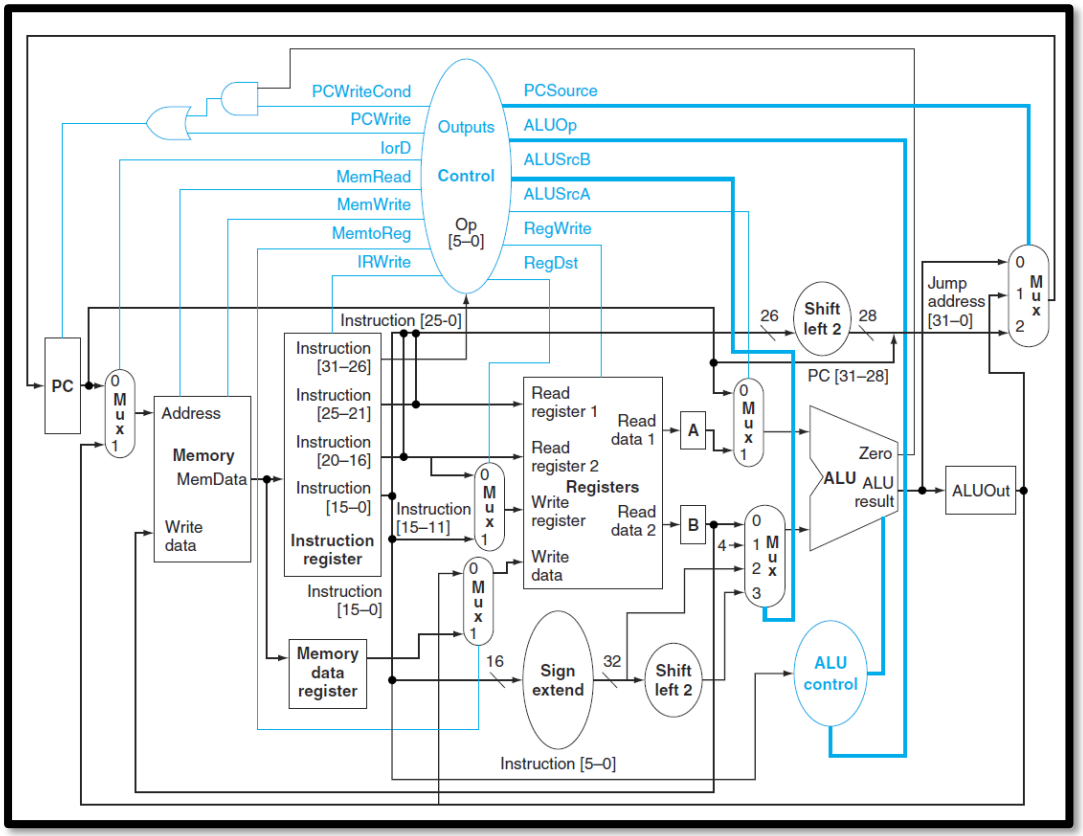
MIPS Multicycle CPU Datapath



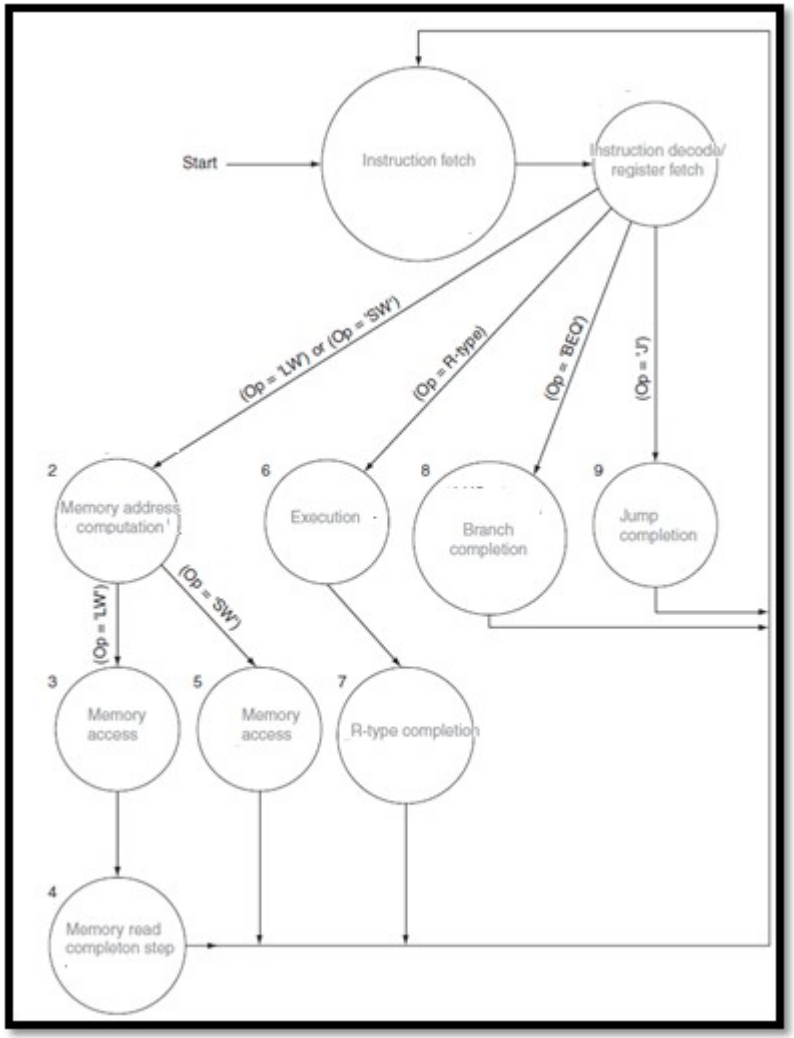
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



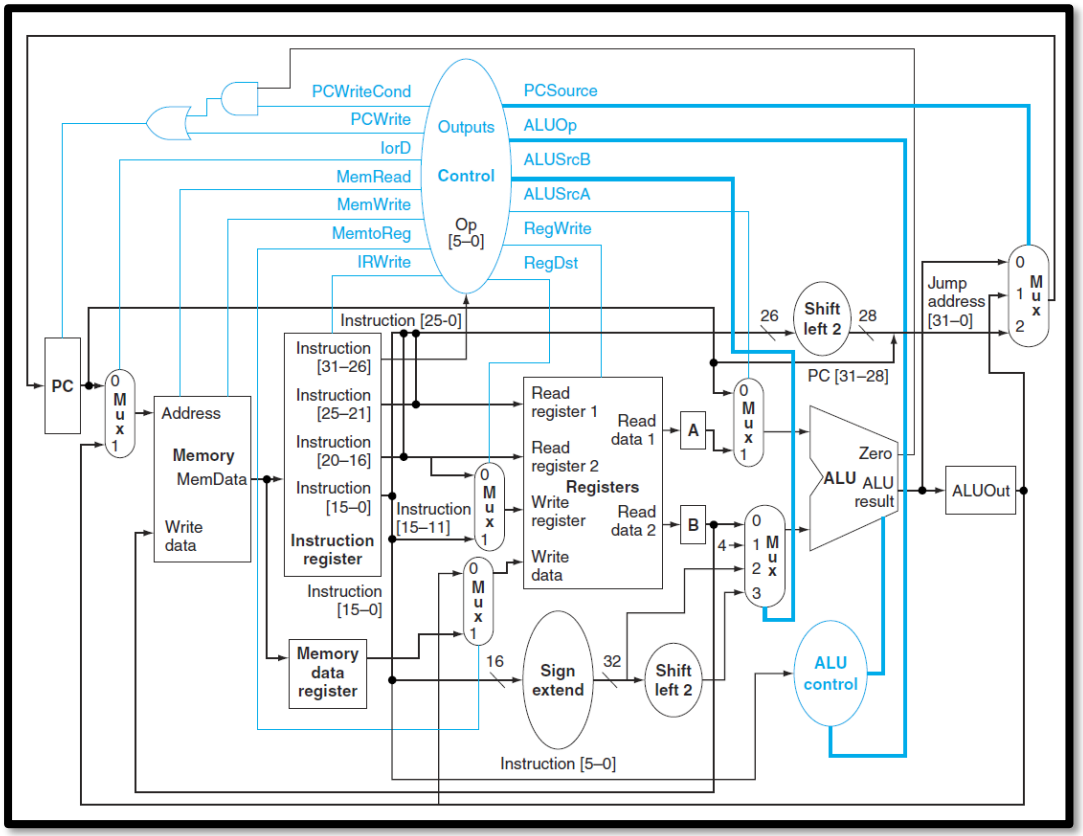
MIPS Multicycle CPU Datapath



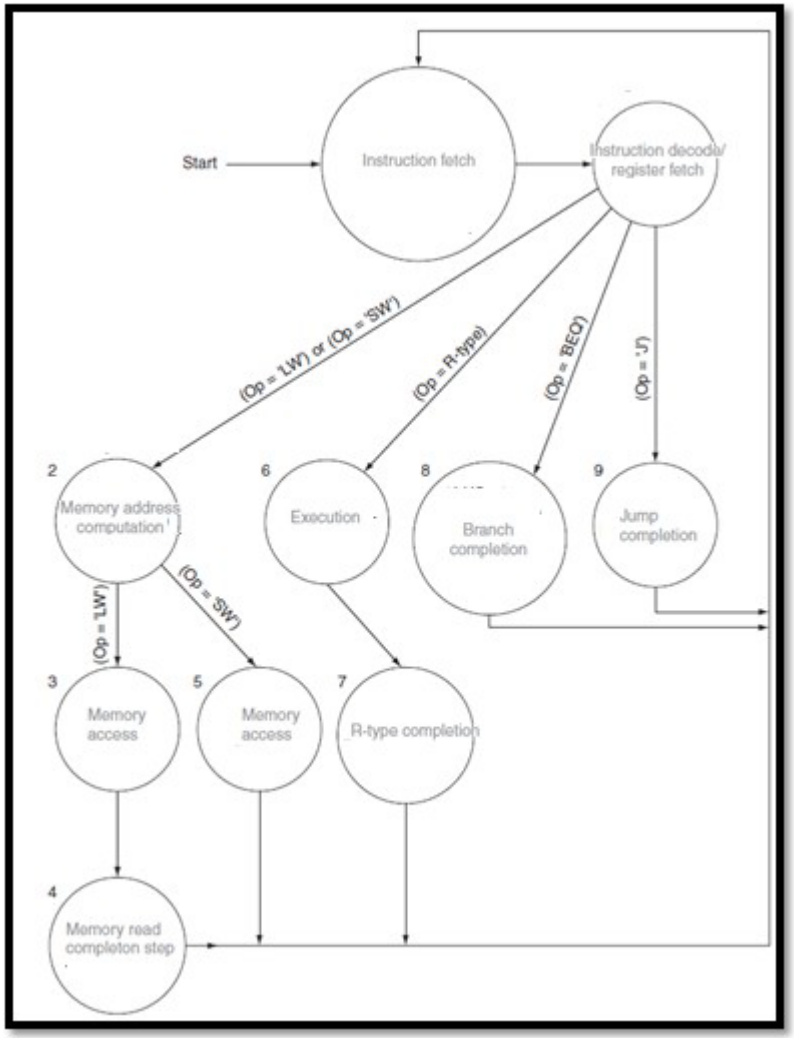
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM

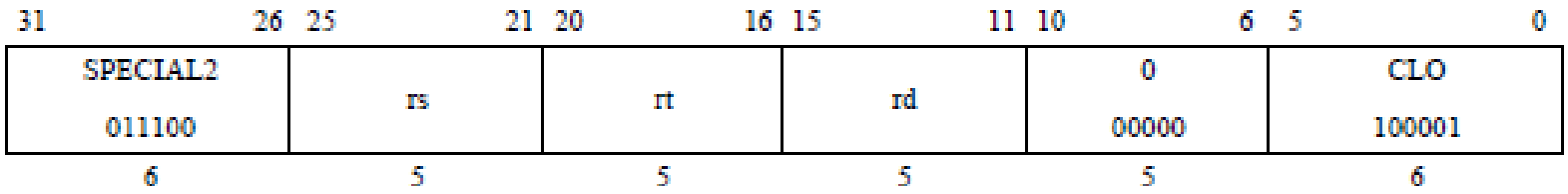


MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller

CLO



Format: `CLO rd, rs` MIPS32

Purpose:

To Count the number of leading ones in a word

Description: `rd ← count_leading_ones rs`

Bits 31..0 of GPR *rs* are scanned from most significant to least significant bit. The number of leading ones is counted and the result is written to GPR *rd*. If all of bits 31..0 were set in GPR *rs*, the result written to GPR *rd* is 32.

MARS – MIPS Simulator

